

Numerical Analysis Homework 1

Zhuqingyu 3230105823

(Mathematics and Applied Mathematics 2301), Zhejiang University

Due time: September 27, 2025

Theoretical Questions

Question 1

Consider the bisection method starting with the initial interval $[1.5, 3.5]$. In the following questions “the interval” refers to the bisection interval whose width changes across different loops.

- (a) What is the width of the interval at the n th step?
- (b) What is the supremum of the distance between the root r and the midpoint of the interval?

Solution:

- (a) Since $h_n = \frac{h_{n-1}}{2} = \frac{h_0}{2^n} = \frac{b_0 - a_0}{2^n} = \frac{3.5 - 1.5}{2^n} = \frac{1}{2^{n-1}}$.
- (b) Since the root $\alpha \in [a_n, b_n]$ and $b_n - a_n = h_n = \frac{1}{2^{n-1}}$, hence $|c_n - \alpha| \leq \frac{1}{2}(b_n - a_n) = \frac{1}{2^n}$.

Question 2

In using the bisection algorithm with its initial interval as $[a_0, b_0]$ with $a_0 > 0$, we want to determine the root with its *relative* error no greater than ϵ . Prove that this goal of accuracy is guaranteed by the following choice of the number of steps,

$$n \geq \frac{\log(b_0 - a_0) - \log \epsilon - \log a_0}{\log 2} - 1.$$

Solution: To make sure that $\frac{|c_n - \alpha|}{\alpha} < \epsilon$, since $\frac{|c_n - \alpha|}{|\alpha|} \leq \frac{|c_n - \alpha|}{|a_n|} \leq \frac{|c_n - \alpha|}{|a_0|} \leq \frac{b_0 - a_0}{2^{n+1}a_0} < \epsilon$, which implies that

$$n \geq \frac{\log(b_0 - a_0) - \log \epsilon - \log a_0}{\log 2} - 1.$$

Question 3

Perform four iterations of Newton's method for the polynomial equation $p(x) = 4x^3 - 2x^2 + 3 = 0$ with the starting point $x_0 = -1$. Use a hand calculator and organize results of the iterations in a table.

Solution: $p(x) = 4x^3 - 2x^2 + 3$ has the derivative $p'(x) = 12x^2 - 4x$. use the initial guess $x_0 = -1$ and $x_{n+1} = x_n - \frac{p(x_n)}{p'(x_n)}$, we have

Table 1: Newton's Method Iterations

Iteration	x_n	$p(x_n)$	$p'(x_n)$	x_{n+1}
0	-1.000000	-3.000000	16.000000	-0.812500
1	-0.812500	-0.465820	11.171875	-0.770804
2	-0.770804	-0.020138	10.212886	-0.768832
3	-0.768832	-0.000043	10.168568	-0.768828

*Electronic address: 3230105823@zju.edu.cn

Question 4

Consider a variation of Newton's method in which only the derivative at x_0 is used,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_0)}.$$

Find C and s such that

$$e_{n+1} = Ce_n^s,$$

where e_n is the error of Newton's method at step n , s is a constant, and C may depend on x_n , the true solution α , and the derivative of the function f .

Solution: Let $g(x) = x - \frac{f(x)}{f'(x_0)}$, then we have $g'(x) = 1 - \frac{f'(x)}{f'(x_0)}$ and $g(\alpha) = \alpha$, so when x_0 is close to α (which is a root of f), i.e. $\exists \delta > 0, \forall x \in [\alpha - \delta, \alpha + \delta], f'(x) \neq 0$, take $x_0 \in [\alpha - \delta, \alpha + \delta]$

$$x_{n+1} - \alpha = g(x_n) - g(\alpha) = \left(1 - \frac{f'(\eta_n)}{f'(x_0)}\right)(x_n - \alpha),$$

where η_n is between x_n and α . Thus $C = 1 - \frac{f'(\eta_n)}{f'(x_0)}$ and $s = 1$.

Question 5

Within $(-\frac{\pi}{2}, \frac{\pi}{2})$, will the iteration $x_{n+1} = \tan^{-1} x_n$ converge?

Solution: The iteration $x_{n+1} = \arctan(x_n)$ converges:

Define $h(x) = \arctan x - x$, $h'(x) = \frac{1}{1+x^2} - 1 < 0, \forall x \in (-\pi/2, \pi/2)$, so $h(x)$ is decreasing. Since $h(0) = 0$, hence when $x > 0$, $h(x) < 0$, when $x < 0$, $h(x) > 0$. If $x_0 > 0$, by induction we have $x_{n+1} - x_n = \arctan(x_n) - x_n < 0$, so $0 < x_{n+1} < x_n$. If $x_0 < 0$, by induction we have $x_{n+1} - x_n = \arctan(x_n) - x_n > 0$, so $x_n < x_{n+1} < 0$. Using monotonic sequence theorem, x_n converges.

Assume that x_n converges to β , since $\arctan(x)$ is continuous, we have $\beta = \arctan(\beta)$, so $\beta = 0$, $x_n \rightarrow 0$ as $n \rightarrow \infty$.

Question 6

Let $p > 1$. What is the value of the following continued fraction?

$$x = \frac{1}{p + \frac{1}{p + \frac{1}{p + \dots}}}$$

Prove that the sequence of values converges. (Hint: this can be interpreted as $x = \lim_{n \rightarrow \infty} x_n$, where $x_1 = \frac{1}{p}$, $x_2 = \frac{1}{p + \frac{1}{p}}$, $x_3 = \frac{1}{p + \frac{1}{p + \frac{1}{p}}}$, ..., and so forth. Formulate x as a fixed point of some function.)

Solution: $x_{n+1} = \frac{1}{p + x_n}$ and $x_n \in (0, 1/p)$. Define $g(x) = \frac{1}{p+x}$, $g'(x) = \frac{-1}{(p+x)^2}$, $|g'(x)| < 1$ for all $x \in (0, 1/p)$, so $g(x)$ is a contraction mapping. Thus $x_n \rightarrow \alpha$ as $n \rightarrow \infty$, where α is a fixed point of g . Solving $\alpha = \frac{1}{p+\alpha}$ gives $\alpha = \frac{-p + \sqrt{p^2 + 4}}{2}$.

Question 7

What happens in problem 2 if $a_0 < 0 < b_0$? Derive an inequality of the number of steps similar to that in problem 2. In this case, is the relative error still an appropriate measure?

Solution: Consider example $y = \arctan x - \delta$, $\alpha_\delta = \tan(\delta)$, take small δ or let $\delta = 0$, relative error $\frac{|x_n - \alpha_\delta|}{\alpha_\delta}$ can't measure the accuracy of the iteration, so we need to use absolute error $|x_n - \alpha_\delta|$. $|x_n - \alpha| \leq \frac{1}{2^{n+1}}(b_0 - a_0) \leq \epsilon \implies n > \frac{\log(b_0 - a_0) - \log \epsilon}{\log 2} - 1$

Question 8

(*) Consider solving $f(x) = 0$ ($f \in \mathcal{C}^{k+1}$) by Newton's method with the starting point x_0 close to a root of multiplicity k . Note that α is a zero of multiplicity k of the function f iff

$$f^{(k)}(\alpha) \neq 0; \quad \forall i < k, \quad f^{(i)}(\alpha) = 0.$$

- How can a multiple zero be detected by examining the behavior of the points $(x_n, f(x_n))$?
- Prove that if r is a zero of multiplicity k of the function f , then quadratic convergence in Newton's iteration will be restored by making this modification:

$$x_{n+1} = x_n - k \frac{f(x_n)}{f'(x_n)}.$$

Solution:

- (a) A multiple zero can be detected by observing that the convergence of Newton's method becomes linear rather than quadratic when approaching a multiple root.
- (b) *Proof.* Since $f \in C^{k+1}$ and α is a root of multiplicity k , we have $f' \in C^k$, $f'' \in C^{k-1}$. Then:

$$\begin{aligned} f(x) &= (x - \alpha)^k \cdot \frac{f^{(k)}(\alpha)}{k!} + O((x - \alpha)^{k+1}) \\ f'(x) &= \frac{f^{(k)}(\alpha)}{(k-1)!} (x - \alpha)^{k-1} + O((x - \alpha)^k) \\ f''(x) &= \frac{f^{(k)}(\alpha)}{(k-2)!} (x - \alpha)^{k-2} + O((x - \alpha)^{k-1}) \end{aligned}$$

We prove that Newton's method has linear convergence. Define:

$$h(x) = x - \frac{f(x)}{f'(x)}$$

Then:

$$h'(x) = 1 - \frac{1}{(f'(x))^2} \cdot [(f'(x))^2 - f(x) \cdot f''(x)] = \frac{f(x)f''(x)}{(f'(x))^2}$$

Note that $h'(x)$ is not defined at $x = \alpha$ since $f'(\alpha) = 0 = f(\alpha)$. However, we can compute the limit as $x \rightarrow \alpha$:

$$\begin{aligned} h'(x) &= \frac{\left((x - \alpha)^k \cdot \frac{f^{(k)}(\alpha)}{k!} + o((x - \alpha)^k) \right) \cdot \left(\frac{f^{(k)}(\alpha)}{(k-2)!} (x - \alpha)^{k-2} + o((x - \alpha)^{k-2}) \right)}{\left(\frac{f^{(k)}(\alpha)}{(k-1)!} (x - \alpha)^{k-1} + o((x - \alpha)^{k-1}) \right)^2} \\ &= \frac{\frac{f^{(k)}(\alpha)^2}{k!(k-2)!} (x - \alpha)^{2k-2} + o((x - \alpha)^{2k-2})}{\frac{f^{(k)}(\alpha)^2}{((k-1)!)^2} (x - \alpha)^{2k-2} + o((x - \alpha)^{2k-2})} \\ &\rightarrow \frac{(k-1)!^2}{k!(k-2)!} = \frac{((k-1)!)^2}{k!(k-2)!} = \frac{k-1}{k} = 1 - \frac{1}{k} \quad (k \geq 2) \end{aligned}$$

So we define $h'(\alpha) = 1 - \frac{1}{k}$ and $h \in C^1$ near α , with $|h'(\alpha)| < 1$. Therefore, Newton's method has linear convergence, and the sequence $(x_n, f(x_n))$ will converge slower than expected.

Define the modified iteration function:

$$r(x) = x - k \cdot \frac{f(x)}{f'(x)} = k \left(x - \frac{f(x)}{f'(x)} \right) - (k-1)x = kh(x) - (k-1)x$$

Then:

$$r'(\alpha) = k \cdot \left(1 - \frac{1}{k} \right) - (k-1) = (k-1) - (k-1) = 0$$

Using Corollary 1.42 [1], we know that at least quadratic convergence will be restored. □

References

- [1] Zhang Qinghai. *Notes on Numerical Analysis and Numerical Methods for Differential Equations*. 2025.